

ROADOCC: LEARNING WHEN TO PERSIST, TRANSPORT, OR REFRESH MEMORY FOR ROADSIDE OCCUPANCY PREDICTION

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ABSTRACT

3D semantic occupancy prediction has advanced rapidly for ego-centric driving, while camera-only roadside occupancy remains underexplored. Fixed cameras make history valuable for sparse or occluded traffic, but coordinate alignment corrects only sensor geometry and leaves moving-object evidence at stale locations. Flow-based fusion can propose a historical address, yet cannot determine whether the retrieved feature provides supported evidence. We therefore formulate temporal memory as supervised soft routing among evidence sources before fusion. In this work, we present RoadOcc, in which each selected voxel predicts whether to *persist* rigidly aligned memory, *transport* memory from a velocity-addressed location, or *refresh* it with current evidence. Dynamic-Aware Cross-Attention (DCA) locates motion-sensitive candidates, Multi-scale Voxel Velocity Estimation (VVE) estimates their transport addresses, and state-conditioned Velocity-Guided Dynamic Sparse Fusion (VDSF) predicts a three-way source-routing distribution over these candidates. This connected design determines where temporal reasoning is needed, where history should be read, and whether it should be used. On InfraOcc, RoadOcc achieves 65.29 mIoU and 32.37 dynamic mIoU, outperforming STCOcc by 4.44 and 4.71 points, respectively, while reducing Direct MAVE from 1.946 to 1.669. Controlled transfer and component ablations further support the routing design. Additional implementation and evaluation details are provided in the supplementary material.

1 INTRODUCTION

Semantic occupancy prediction reconstructs free space, static layout, and traffic participants in a common voxel volume. Camera-only methods have made this representation practical through sparse voxel queries, BEV lifting, tri-plane features, and unified panoptic volumes (Cao & de Charette, 2022; Li et al., 2023a; Wang et al., 2023; Tian et al., 2023; Li et al., 2025b; Huang et al., 2023; Wei et al., 2023; Tang et al., 2024; Huang et al., 2024b; Wang et al., 2024; Li et al., 2025a). However, these methods are predominantly developed for moving platforms, where each observation is interpreted in an ego-centric frame. A roadside sensor instead repeatedly observes the same scene, producing a persistent scaffold overlaid by sparse, short-lived dynamic events. This makes temporal memory valuable but turns physically correct reuse into a central challenge.

The same history can provide either valid support or stale evidence. Rigidly aligned memory preserves static layout and can recover weak or occluded vehicle boundaries, but it does not relocate an object that moved within the fixed coordinate system. Reading such memory from the former footprint can blur a dynamic boundary, duplicate an object, or attach it to nearby static layout. Discarding history avoids these errors but also removes useful support. The challenge is therefore not simply whether to use history, but which form of history is valid at each location.

Recent temporal methods demonstrate the value of multi-frame context for occupancy, flow, and 3D perception (Ma et al., 2024; Zheng et al., 2024; Wang et al., 2025; Chen et al., 2025a; Liao et al., 2025; Liu et al., 2024; Huang & Huang, 2022; Chang et al., 2024; Leng et al., 2025). Existing designs already incorporate temporal memory, motion alignment, and adaptive fusion. However, they largely formulate history as a retrieved feature whose confidence is adjusted after alignment. This

leaves one unresolved question: whether the feature at a flow-proposed address actually supports the current voxel. A stationary object may reuse rigid memory, a moving object may retrieve memory near $\mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})$, and an appearing, occluded, or historically unsupported object must instead rely on current evidence. Continuous feature weighting can represent these alternatives, but existing formulations typically lack a direct training signal for when transported history must be rejected. We therefore formulate temporal occupancy as supervised evidence-source routing. *Should each selected voxel Persist, Transport, or Refresh?* Persist reads the rigidly aligned address, Transport reads a velocity-compensated address, and Refresh uses current image evidence. The core principle is not to make flow “stronger”: flow supplies a candidate source address, whereas P/T/R decides whether rigid history, transported history, or no history is admissible for the current voxel. This decision is directly supervisable and exposes which source supports each prediction.

We instantiate this principle in RoadOcc as a staged routing process. Dynamic-Aware Cross-Attention (DCA) first uses current semantics to identify the voxels that require temporal reasoning. Restricted to these candidates, Multi-scale Voxel Velocity Estimation (VVE) estimates where their historical support should be retrieved. State-conditioned Velocity-Guided Dynamic Sparse Fusion (VDSF) then uses the resulting addresses to predict a soft distribution over rigid memory, transported memory, and current evidence. Because each stage resolves the uncertainty required by the next, the temporal source choice becomes explicit and trainable. Source-aligned P/T/R targets and a multi-scale route objective provide direct supervision for the entire chain. Our evaluation tests the same mechanism rather than treating velocity as an isolated auxiliary task. The InfraOcc benchmark measures predictive accuracy, while dynamic-first temporal diagnostics test whether gains coincide with improved support coverage without sacrificing static accuracy. Figure 1 grounds the three evidence-source alternatives in one real roadside sequence. Our contributions are summarized below.

- We propose RoadOcc, which formulates roadside temporal occupancy as Persist–Transport–Refresh routing issue. The router decides whether a flow-proposed historical source is admissible, separating reusable history from evidence that must be relocated or discarded.
- We realize this principle through a staged DCA–VVE–VDSF chain. DCA selects where temporal reasoning is needed, enabling VVE to estimate motion-compensated historical addresses that VDSF uses to softly route rigid, transported, or current evidence.
- We evaluate the source-routing hypothesis with dynamic-first occupancy, flow-advected next-frame consistency, velocity, and support-recall diagnostics. Controlled ablations isolate address estimation from the decision to adopt or refresh historical evidence.

2 RELATED WORK

2.1 CAMERA OCCUPANCY AND ROADSIDE PERCEPTION

Camera occupancy methods lift multi-view image evidence into volumetric representations. MonoScene Cao & de Charette (2022) and VoxFormer Li et al. (2023a) establish image-

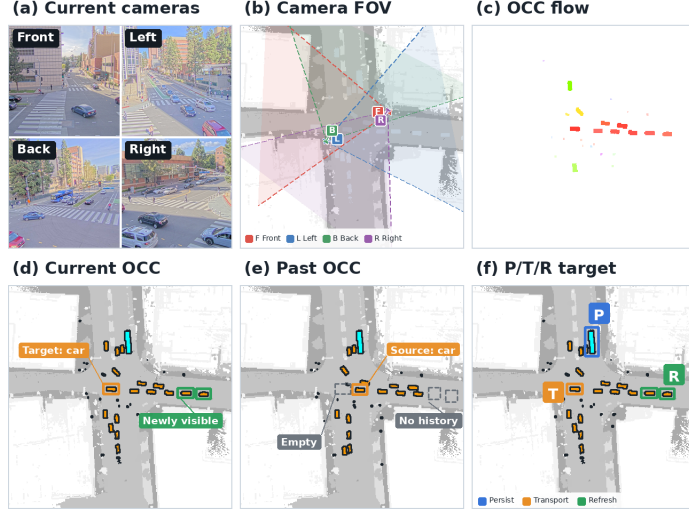


Figure 1: **Real-scene evidence for explicit source routing.** The top row shows current views, camera FOVs, and current-aligned occupancy flow (a–c). The bottom row shows current/past occupancy and P/T/R targets (d–f), illustrating a persistent bus, transported car, and newly visible cars assigned Refresh.

based semantic scene completion, while BEVDet, BEVDepth, and BEVFormer improve multi-view lifting and BEV reasoning (Huang et al., 2021; Li et al., 2023b; 2025b). Occupancy-specific designs span tri-plane, dense, sparse, Gaussian, and forward-backward representations (Huang et al., 2023; Wei et al., 2023; Zhang et al., 2023; Tang et al., 2024; Huang et al., 2024b; Li et al., 2023c), as well as panoptic and instance-aware prediction (Wang et al., 2024; Li et al., 2025a). Unlike MC-BEVRO’s roadside BEV occupancy monitoring (Vaghela et al., 2025), RoadOcc routes evidence for class-aware 3D occupancy.

InfraOcc provides a roadside occupancy benchmark emphasizing persistent infrastructure (Yang et al., 2026). Figure 2 reveals its temporal asymmetry. Fixed roadway structure provides broad, stable support, while dynamic participants form sparse, short-lived trails. This asymmetry favors retaining static context while selectively relocating moving evidence.

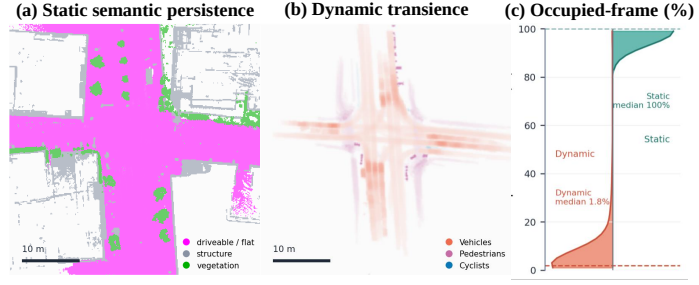


Figure 2: Static-to-dynamic persistence in InfraOcc.

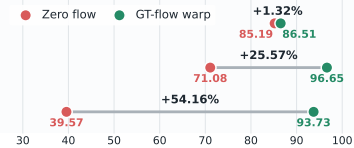
2.2 TEMPORAL OCCUPANCY, MOTION, AND ALIGNMENT

Temporal occupancy methods maintain or forecast scene state, sometimes jointly with flow. Cam4DOcc, OccWorld, UniOcc, ALOcc, STCOcc, and Let Occ Flow model dynamics alongside volumetric semantics (Ma et al., 2024; Zheng et al., 2024; Wang et al., 2025; Chen et al., 2025a; Liao et al., 2025; Liu et al., 2024; Zheng et al., 2026). ST-Occ conditions memory on uncertainty and dynamics, while GDFusion organizes scene, motion-calibration, and geometric cues for temporal fusion (Leng et al., 2025; Chen et al., 2025b). Recurrent BEV models also maintain aligned memory (Huang & Huang, 2022; Chang et al., 2024), while SelfOcc and ForecastOcc learn from video or predict future occupancy (Huang et al., 2024a; Mohan et al., 2026). These methods establish motion-aware alignment and adaptive fusion. RoadOcc focuses on the subsequent source-validity decision by directly supervising a mutually exclusive choice among rigid history, transported history, and current evidence. Its next-frame score evaluates the resulting motion field rather than a forecasting head.

Table 1: **Generated GT-flow target at $t + 0.5$ s.** Zero-flow and GT-flow transport share the inter-frame transform and differ only by object displacement. Same/Dyn./Occ. denote unchanged semantics, dynamic class, and occupied class. Values are percentages and N counts valid queries in millions.

Speed	N (M)	Zero-flow transport			GT-flow warp			Δ Same
		Same	Dyn.	Occ.	Same	Dyn.	Occ.	
All	11.83	72.93	78.86	90.92	89.21	95.79	99.38	+16.28
Slow (< 2 m/s)	7.87	85.19	94.08	98.80	86.51	96.25	99.70	+1.32
Med. (2–5 m/s)	1.16	71.08	71.38	87.50	96.65	97.64	99.53	+25.57
Fast (≥ 5 m/s)	2.80	39.57	39.59	70.41	93.73	93.75	98.43	+54.16

Figure 3: **Motion compensation restores temporal consistency.** GT flow preserves dynamic support across speed ranges, whereas rigid retrieval becomes increasingly misaligned as speed grows.



Flow-based alignment is closest to this problem. AsyncBEV aligns asynchronous cross-modal observations with BEV feature flow (Wang et al., 2026). In the fixed roadside setting, Table 1 and Figure 3 show that GT-flow warping improves same-class agreement by 1.32 points for slow objects, but by 25.57 and 54.16 points for medium and fast objects. This speed-dependent gap favors selective motion compensation over a uniform temporal warp.

3 METHOD

3.1 PRELIMINARY ANALYSIS

Let $\mathcal{I}_t = \{I_t^n\}_{n=1}^4$ denote the synchronized roadside images at time t . RoadOcc predicts semantic occupancy $\mathbf{y}_t$ over a voxel domain Ω and a planar velocity field $\mathbf{v}_t : \Omega \rightarrow \mathbb{R}^2$. Among the 18 occu-

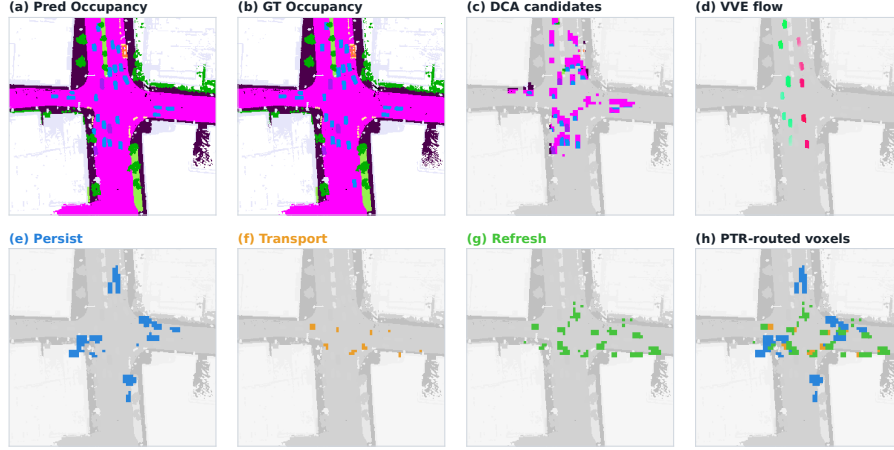


Figure 4: **RoadOcc execution trace.** Current occupancy evidence identifies DCA candidates (a–c), VVE estimates their motion-aware historical addresses (d), and VDSF separates routed voxels into Persist, Transport, and Refresh evidence (e–h).

pancy states, the dynamic set is $\mathcal{C}_{\text{dyn}} = \{\text{bicycle, bus, car, motorcycle, pedestrian, truck}\}$. Velocity is expressed in the common roadside horizontal frame.

Rigid alignment removes camera motion but not object motion. Under a locally constant velocity approximation, history supporting a current voxel $\mathbf{x}$ is read from

$$\tilde{\mathbf{h}}_{t-\Delta t}(\mathbf{x}) = \text{Sample}(\bar{\mathbf{h}}_{t-\Delta t}, \mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})), \quad (1)$$

where $\bar{\mathbf{h}}$ is rigidly aligned history and Sample is trilinear interpolation. The three candidates are rigid history at $\mathbf{x}$ (Persist), flow-addressed history at $\mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})$ (Transport), and current evidence (Refresh). Equation 1 specifies only an address. VDSF still decides whether the sampled source has class-consistent historical support.

3.2 OVERVIEW

RoadOcc decodes $\{\mathbf{V}_t^s\}_{s \in \{1/8, 1/4, 1/2\}}$ from coarse to fine. At each scale, DCA revisits motion-sensitive image evidence, VVE proposes its historical address, and VDSF routes rigid, transported, or current evidence. Estimating the address before source admissibility lets VDSF reject unsupported memory. The fused state is upsampled to the next scale, yielding full-resolution occupancy and velocity. Figures 4 and 5 show this sequence and its progressive output.

3.3 DYNAMIC-AWARE CROSS-ATTENTION (DCA)

Dynamic objects occupy few voxels, so uniformly repeating image cross-attention spends most computation on stable structure. DCA instead derives a candidate map after the first nonempty-guided image update. Let $\pi_{t,\text{cur}}^s$ and $\pi_{t,\text{sta}}^s$ denote the provisional distributions produced by the current and static prediction heads after the first image update, and let $\mathcal{C}_{\text{sta}}^+$ contain the static classes and free space. DCA uses two complementary cues. The first is the mean semantic discrepancy $\delta_{t,\text{sta}}^s$ between the two distributions over $\mathcal{C}_{\text{sta}}^+$. The second is the current dynamic probability $p_{t,\text{dyn}}^s$. Their normalized maximum forms the candidate map

$$c_t^s(\mathbf{x}) = \text{Norm}(\max\{\delta_{t,\text{sta}}^s(\mathbf{x}), p_{t,\text{dyn}}^s(\mathbf{x})\}). \quad (2)$$

Here Norm divides by the spatial maximum within each sample, with the denominator clamped to 10^{-6} . The discrepancy cue recovers locations whose current evidence departs from the static hypothesis, while the direct cue retains already visible actors. Taking their maximum is important because an early semantic miss can still be recovered from change evidence instead of being removed from subsequent reasoning.

The candidate map controls a second image cross-attention rather than directly reweighting the voxel feature. Following projected deformable attention (Li et al., 2025b) and occupancy-aware reference

sampling (Liao et al., 2025), let $\mathbf{q}_{t,\mathbf{x}}^s$ be the BEV query at $\mathbf{x}$, $\mathcal{R}_n(\mathbf{x})$ its visible height anchors in camera n , and $\mathbf{F}_t^n$ the corresponding image feature. DCA retains $\mathcal{R}_{n,c}^s(\mathbf{x}) = \{\mathbf{r} \in \mathcal{R}_n(\mathbf{x}) : u_{\mathbf{r}} < c_t^s(\mathbf{r})\}$ and updates the query as

$$\mathbf{q}_{t,\mathbf{x}}^{s,+} = \mathbf{q}_{t,\mathbf{x}}^s + \mathbf{W}_o \frac{1}{|\mathcal{V}_{\mathbf{x}}|} \sum_{n \in \mathcal{V}_{\mathbf{x}}} \sum_{\mathbf{r} \in \mathcal{R}_{n,c}^s(\mathbf{x})} \beta_{n,\mathbf{r}}^s \mathcal{A}_{\text{def}}(\mathbf{q}_{t,\mathbf{x}}^s, \mathcal{P}_n(\mathbf{r}), \mathbf{F}_t^n), \quad (3)$$

where $\mathbf{r} = (\mathbf{x}, z_{\mathbf{r}})$ indexes a voxel-height score shared across cameras, $\mathcal{V}_{\mathbf{x}}$ contains valid cameras, and $\mathcal{P}_n$ projects anchors. Depth consistency $\beta_{n,\mathbf{r}}^s$ multiplies normalized deformable weights before camera averaging. During training, $u_{\mathbf{r}}$ is sampled from a normal distribution with mean 0.5 and standard deviation 1, truncated to $[0, 1]$. At inference, it is fixed to 0.5. The stochastic rule exposes uncertain boundary voxels during training, while deterministic thresholding keeps inference stable. Queries without retained support follow the residual path unchanged.

The second update supplies renewed current evidence to VVE, while the same map allocates VDSF tokens. DCA localizes where evidence is needed. VVE determines its temporal source.

3.4 MULTI-SCALE VOXEL VELOCITY ESTIMATION (VVE)

DCA refines current evidence but cannot align moving objects with displaced memory. VVE estimates this displacement from DCA features and rigidly aligned history at each decoder scale. Let $\pi_t^s(\mathbf{x})$ be the current occupancy distribution. From it, VVE computes the dynamic support $d_t^s = \sum_{c \in \mathcal{C}_{\text{dyn}}} \pi_{t,c}^s$ and the nonempty support $e_t^s = 1 - \pi_{t,\text{free}}^s$. These cues serve different purposes. d_t^s suppresses static structure when the DCA-refined voxel volume is projected into the current BEV feature $\mathbf{B}_t^s$, whereas e_t^s is retained for temporal token selection. Because roadside motion is predominantly horizontal, this projection concentrates the matching problem in BEV without discarding the voxel-wise output structure. An analogous support-weighted projection of aligned history gives $\bar{\mathbf{B}}_{t-\Delta t}^s$.

VVE next compares each current BEV location with a small neighborhood in the aligned history, producing the local correspondence volume

$$\mathbf{K}_t^s(\mathbf{x}, \delta) = \left\langle \frac{\mathbf{B}_t^s(\mathbf{x})}{\|\mathbf{B}_t^s(\mathbf{x})\|_2}, \frac{\bar{\mathbf{B}}_{t-\Delta t}^s(\mathbf{x} + \delta)}{\|\bar{\mathbf{B}}_{t-\Delta t}^s(\mathbf{x} + \delta)\|_2} \right\rangle, \quad \delta \in [-r, r]^2, \quad (4)$$

where $r = 2$ yields a 5×5 neighborhood and both norms use an $\epsilon = 10^{-6}$ floor. With current/history features, $\mathbf{K}$ decodes a two-channel metric-velocity residual for locally distinct motion.

VVE estimates velocity recursively rather than solving each resolution in isolation. Let s^- denote the preceding coarser stage and set the upsampled prior $\mathbf{v}_{t,\uparrow}^{s^-}$ to zero at $1/8$ scale and to $\text{Up}_s(\hat{\mathbf{v}}_t^{s^-})$ otherwise. Current appearance and its semantic embedding $\mathbf{S}_t^s$ predict a current-only correction $\Delta \mathbf{v}_{t,\text{cur}}^s$, while aligned history and the correspondence volume predict $\Delta \mathbf{v}_{t,\text{hist}}^s$. VVE combines them as

$$\hat{\mathbf{v}}_t^s = \mathbf{v}_{t,\uparrow}^{s^-} + (1 - \alpha_t^s) \Delta \mathbf{v}_{t,\text{cur}}^s + \alpha_t^s \Delta \mathbf{v}_{t,\text{hist}}^s. \quad (5)$$

Here α_t^s blends by current dynamic support and valid history, and is zero at sequence starts. Coarse stages capture low-resolution motion structure and fine stages refine boundaries. The final BEV field is lifted across height to provide addresses. VDSF, rather than VVE, decides whether to use the retrieved feature.

3.5 VELOCITY-GUIDED DYNAMIC SPARSE FUSION (VDSF)

Velocity-guided sparse fusion. Applying the full FIFO memory after DCA and VVE would mostly repeat static context. VDSF makes gather–fuse–scatter (Liao et al., 2025) motion aware by combining DCA guidance with nonempty support under a fixed token budget

$$\mathbf{g}_t^s = \text{clip}(c_t^s + \eta e_t^s, 0, 1), \quad \mathcal{X}_t^s = \text{TopK}(\mathbf{g}_t^s), \quad (6)$$

where η balances change evidence and occupied support. The selected set $\mathcal{X}_t^s$ receives full stage-local history. Its complement uses a short rigid-history path, preserving background context at lower cost.



Figure 5: **Coarse-to-fine occupancy decoding.** A representative scene is shown. Columns present predictions from 1/8 scale to full resolution and the corresponding ground truth. Successive stages recover road boundaries and traffic participants that remain unresolved at coarse resolution.

Each scale maintains a FIFO queue of fused features and VVE priors, rigidly aligned before retrieval. For $\mathbf{x} \in \mathcal{X}_t^s$, slot age converts velocity into a voxel offset. VDSF gathers one source-consistent inverse-trajectory address as Transport candidate $\mathbf{h}^T$ and the same-coordinate source as Persist candidate $\mathbf{h}^P$.

Persist-Transport-Refresh routing. Velocity identifies where history may be read, but not how that history should be weighted. VDSF therefore retains $\mathbf{h}^P$ and $\mathbf{h}^T$ and adds current evidence $\mathbf{h}^R = \mathbf{q}_t$. During training, the route target is constructed from the current and nearest historical semantic occupancy labels together with the motion target, and assigns each current dynamic voxel according to class-consistent historical support:

$$z_t(\mathbf{x}) = \begin{cases} P, & \|\mathbf{v}_t(\mathbf{x})\| \leq \epsilon \wedge \text{Supp}(\mathbf{x}), \\ T, & \|\mathbf{v}_t(\mathbf{x})\| > \epsilon \wedge \text{Supp}(\mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})), \\ R, & \text{otherwise,} \end{cases} \quad (7)$$

where Supp tests same-class occupancy at the same height within the local correspondence radius. We use $\epsilon = 10^{-3}$ m/s and a one-voxel radius to absorb discretization error. The construction follows the semantic occupancy label space used by InfraOcc and is evaluated at every execution scale using its physical cell size, with a native-grid endpoint for full-resolution supervision. Under this rule, supported stationary voxels Persist, supported moving voxels Transport, and voxels without a supported historical candidate Refresh. The supplement specifies the source-alignment and label-generation procedure.

At each decoder scale, a shared controller combines the current feature, nearest rigidly aligned history, normalized VVE flow, and upsampled coarser P/T/R distribution. It predicts stage-local logits; a lightweight native head upsamples the finest context with full-resolution flow. Softmax gives $\alpha_t^s = (\alpha_P, \alpha_T, \alpha_R)$ and the routed candidate

$$\mathbf{h}_t^{\text{PTR}} = \alpha_P \mathbf{h}^P + \alpha_T \mathbf{h}^T + \alpha_R \mathbf{h}^R, \quad (\alpha_P, \alpha_T, \alpha_R) = \text{softmax}(\psi(\cdot)). \quad (8)$$

P/T/R therefore compete as one route distribution rather than independent weights. Fusion uses the soft probabilities; argmax is reserved for diagnostics. Transport is emphasized only when supported, while conflicting history can fall back to Refresh.

Let $\mathbf{H}_t^{\text{PTR},s}(\mathbf{x})$ concatenate the routed candidates from all valid FIFO slots at scale s . VDSF fuses the temporal stack with the current token and its semantic embedding through a channel MLP

$$\mathbf{q}_{t,\text{fuse}}^s(\mathbf{x}) = \mathbf{q}_t^s(\mathbf{x}) + \Phi_s \left([\mathbf{q}_t^s(\mathbf{x}), \mathbf{H}_t^{\text{PTR},s}(\mathbf{x}), \text{Emb}(\pi_t^s(\mathbf{x}))] \right). \quad (9)$$

The residual is scattered to $\mathcal{X}_t^s$ and joined with the background path. The fused volume and VVE prior update the stage-local queue. Queue lengths, token budgets, source matching, and routing settings appear in supplementary Secs. 3–5.

3.6 TRAINING OBJECTIVE

Training jointly optimizes four functional objectives.

$$\mathcal{L} = \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{sem}} + \lambda_{\text{mot}} \mathcal{L}_{\text{mot}} + \lambda_{\text{route}} \mathcal{L}_{\text{route}}. \quad (10)$$

$\mathcal{L}_{\text{depth}}$ supervises the image view transform with camera-depth targets. $\mathcal{L}_{\text{sem}}$ groups full-resolution and multi-scale semantic occupancy losses together with the static/dynamic and DCA-candidate

Table 2: Main InfraOcc occupancy results. InfraOcc camera-only protocol.

Method	gIoU				mIoU																	
		all	dyn	sta																		
BEVDet '21	68.62	45.14	21.19	63.11	9.23	42.75	38.04	9.68	15.01	12.45	42.88	70.36	53.77	72.04	68.07	73.09	64.28	60.38	94.87			
BEVFormer '22	81.72	47.76	10.73	75.53	7.97	20.39	17.65	3.47	8.62	6.30	42.47	76.26	72.83	85.19	81.95	88.95	79.52	77.05	97.56			
BEVDepth '23	71.89	46.81	19.64	67.19	14.84	33.87	38.74	3.42	16.86	10.14	48.73	71.05	62.37	73.31	69.32	76.07	70.51	66.15	95.54			
TPVFormer '23	73.80	43.09	4.36	72.13	0.10	3.69	13.80	0.41	6.86	1.33	42.86	75.29	69.69	80.61	77.62	86.63	72.61	71.70	96.12			
SurroundOcc '23	83.78	50.77	11.37	80.32	12.66	21.90	18.54	2.06	9.50	3.55	65.09	79.04	72.70	86.38	83.99	91.10	83.16	81.09	97.80			
CONet '23	78.56	49.25	18.93	72.00	3.06	37.24	39.47	4.42	17.64	11.72	45.01	75.31	64.94	84.29	79.45	84.27	72.96	69.79	97.08			
ALOCC '25	75.66	43.31	5.86	71.40	8.41	4.04	8.52	0.00	4.48	9.71	63.90	68.97	63.40	90.09	77.77	78.16	65.07	63.85	97.12			
Let Occ Flow '24	87.18	51.38	8.78	83.34	11.27	11.53	14.07	4.34	8.12	3.34	62.06	75.06	85.56	92.55	85.63	92.98	87.10	85.74	98.56			
SparseOcc '24	71.61	45.42	9.41	72.42	1.50	20.94	24.87	0.00	4.60	4.57	67.68	76.37	61.51	74.67	76.59	88.54	68.79	65.19	96.31			
GaussianFormer '24	77.89	43.63	10.45	68.51	3.13	22.11	23.17	1.29	8.21	4.80	36.70	68.54	55.60	82.59	79.37	74.20	77.85	73.21	96.71			
CRT-Fusion-S '24	88.73	55.94	21.98	81.40	13.02	45.16	38.48	7.34	18.99	8.90	63.53	71.99	74.82	93.38	84.40	91.19	86.39	85.53	98.73			
STCOcc-S '25	78.54	50.71	23.48	71.14	13.43	47.83	39.62	10.21	17.83	11.93	50.18	71.54	58.33	90.30	78.78	84.90	67.72	67.32	97.47			
CRT-Fusion '24	80.35	53.83	27.02	73.94	32.58	46.63	39.49	12.22	20.81	10.41	53.08	72.27	68.15	89.06	79.09	84.48	72.06	73.30	97.68			
STCOcc '25	92.54	60.85	27.66	85.74	27.13	48.78	44.93	10.23	22.85	12.03	68.84	70.93	81.60	95.88	89.41	95.36	91.85	92.04	99.18			
RoadOcc (ours)	94.03	65.29	32.37	89.98	37.60	55.46	49.36	13.01	22.66	16.13	87.66	72.05	86.03	97.85	92.81	96.90	92.91	93.66	99.35			

Table 3: Joint occupancy-velocity comparison. Results are reported for temporal variants under a common protocol. Higher is better except MAVE, which is reported in m/s. Shaded rows isolate the measured change after adding VDSF P/T/R to each baseline.

Method	Occupancy					Velocity	Diagnostics	
	mIoU	Dyn.	Sta.	gIoU	RayIoU	Direct MAVE	TP-MAVE	DSR
CRT-Fusion (NeurIPS 2024)	53.83	27.02	73.94	80.35	48.08	2.227	1.839	57.56
+ VDSF P/T/R	57.15	28.30	78.79	83.46	52.34	1.990	1.958	<u>59.45</u>
Δ	+3.32	+1.28	+4.85	+3.11	+4.26	-0.237	+0.119	+1.89
STCOcc (CVPR 2025)	60.85	27.66	85.74	92.54	55.94	1.946	0.703	57.12
+ VDSF P/T/R	<u>61.81</u>	<u>28.80</u>	<u>86.56</u>	<u>93.59</u>	<u>58.07</u>	<u>1.770</u>	0.648	58.78
Δ	+0.96	+1.14	+0.82	+1.05	+2.13	-0.176	-0.055	+1.66
RoadOcc (ours)	65.29	32.37	89.98	94.03	65.44	1.669	<u>0.661</u>	61.11

auxiliary signals. $\mathcal{L}_{\text{mot}}$ applies class- and speed-balanced robust velocity regression plus local connectivity regularization, directly supervising the Transport address in Eq. 1. $\mathcal{L}_{\text{route}}$ combines cross-entropy with equal P/T/R class weights and soft state overlap on Eq. 7, restricted to valid current dynamic voxels. Missing correspondence remains a Refresh example because rejecting history is part of the task. Thus $\mathcal{L}_{\text{mot}}$ supervises where history is queried, whereas $\mathcal{L}_{\text{route}}$ supervises whether that source is used. Scale-wise weights and schedules appear in supplementary Sec. 3.

4 EXPERIMENTS

Dataset and metrics. We evaluate camera-only roadside occupancy on InfraOcc, derived from V2X-Real (Xiang et al., 2024; Yang et al., 2026). Four synchronized views are mapped to 18 occupancy states. Under the InfraOcc protocol, semantic mIoU excludes construction vehicle, trailer, other flat, and Free. Free IoU is reported separately in the class-wise table. We report semantic and geometric IoU, next-frame dynamic IoU, and motion coverage. Direct MAVE covers all GT dynamic voxels, while TP-MAVE conditions on semantic matches and is paired with recall (DSR). InfraOcc is a concurrent dense fixed-roadside occupancy benchmark; because existing roadside datasets mainly provide object-level labels, we use its camera-only protocol and assess generality through matched transfer and controlled studies, with full benchmark details in the supplement.

Implementation details. RoadOcc uses a ResNet-50 backbone, 256×704 images, and three voxel stages from 0.8 m to 0.4 m over $[-64, 64] \times [-64, 64] \times [-4.8, 1.6]$ m. Frames are 0.5 s apart. AdamW training runs for 24 epochs from 5×10^{-4} with 10^{-2} weight decay and cosine annealing. Comparisons share the voxel range, ignored classes, and evaluator. The -S suffix disables temporal history while retaining the task architecture and training protocol, whereas unmarked temporal variants use their corresponding memory path. Controlled variants keep the backbone, data split, optimizer, schedule, queue reset, and sparse-token configuration fixed unless their designated factor changes. Exact metric, queue, selection, and routing contracts appear in the supplement.

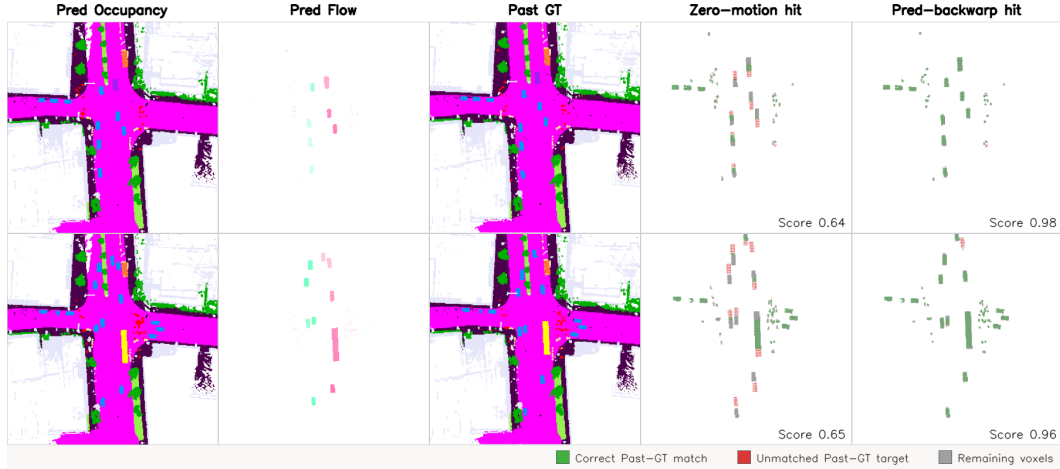


Figure 6: **Velocity-guided historical retrieval.** With queried dynamic targets fixed, each row compares a rigid zero-motion read with the VVE backtrace. Predicted occupancy and flow define addresses in past occupancy; VVE changes unmatched targets (red) into same-class past-GT matches (green). The score is the same-class hit rate within the one-voxel matching radius.

4.1 MAIN RESULTS

InfraOcc occupancy benchmark. Table 2 tests whether the proposed routing design improves occupancy under the InfraOcc evaluator. RoadOcc achieves the strongest result in Table 2 and establishes a new state of the art at 65.29 mIoU, a +4.44 gain over STCOcc. Dynamic mIoU rises by +4.71 to 32.37, while static mIoU rises by +4.24 to 89.98. The joint gain matches the proposed routing mechanism. VVE relocates evidence for moving participants, while P/T/R restricts reuse to supported history. The class-wise results therefore show stronger transient recovery without sacrificing the persistent roadside structure.

Joint occupancy-velocity behavior. Table 3 tests whether these occupancy gains follow the expected routing mechanism. Direct MAVE measures the quality of the VVE address, and DSR measures the dynamic support retained by P/T/R. Relative to STCOcc, RoadOcc lowers Direct MAVE from 1.946 to 1.669 and raises DSR from 57.12 to 61.11. Adding VDSF P/T/R to CRT-Fusion and STCOcc also raises Dyn. by +1.28 and +1.14, respectively, while improving Direct MAVE and DSR. The matching changes show that VDSF P/T/R generalizes to two flow-aware occupancy backbones, linking the occupancy gain to better address correction and source selection. TP-MAVE conditions on correctly matched semantics and must therefore be read with DSR. Its increase on CRT-Fusion accompanies a +1.89 DSR gain, exposing the coverage-conditional-error trade-off rather than conflating a lower conditional error with broader dynamic recovery.

4.2 ABLATION STUDIES

Using a shared backbone and (Liao et al., 2025) as the baseline, we sequentially ablate RoadOcc’s decision chain. DCA selects where temporal evidence is needed, VVE locates its historical source, and P/T/R assesses source admissibility. This staged design separates their contributions from generic temporal fusion. Route-wise diagnostics appear in the supplement.

Overall ablation. Table 4 follows the routing chain. VDSF alone adds only 0.20 mIoU because selecting history without first resolving its location leaves moving evidence misaligned. DCA and VDSF then raise Dyn. by +1.55 and Nxt. by +1.80, showing that current evidence identifies where sparse temporal fusion is useful. With this target set fixed, VVE adds +1.16 mIoU by correcting the historical

Table 4: **Component ablation.**

Base	Design				Output ↑		
	DCA	VVE	VDSF	Ptr	All	Dyn.	Nxt.
✓					60.85	27.66	26.74
✓			✓		61.05	28.01	26.99
✓	✓				61.69	28.55	27.64
✓	✓		✓		63.01	30.10	29.44
✓	✓	✓	✓		64.17	30.97	30.78
✓	✓	✓	✓	✓	65.29	32.37	31.72

address. P/T/R adds a further +1.12 mIoU by deciding whether that source should be reused. The ordered gains align with the three decisions rather than with undifferentiated temporal capacity.

Address correction and source selection. Table 5 holds capacity and token budget fixed. The rigid and VVE reads differ only in the historical address. Their difference raises Dyn. by +0.87, lowers dMAVE from 2.146 to 1.690, and raises DSR by +1.51, directly verifying address correction. With that address fixed, P/T/R adds +1.40 Dyn. and +1.76 DSR. This second transition shows that a corrected address is useful only when the fusion stage can select supported evidence.

P/T/R routing controls. Table 6 holds DCA–VVE–VDSF fixed and isolates source selection. Gate learns unsupervised source weights, State adds P/T/R targets, and Full also uses their distribution for fusion. Across three seeds, the means improve from Gate to State to Full. Full gains +1.40 Dyn., +0.94 Nxt., and +1.76 DSR over None, showing that route supervision and supervised fusion turn a corrected address into a reliable source decision.

P/T/R source-admissibility supervision. P/T/R supervises semantic source validity rather than instance correspondence. Persist and Transport require same-class, height-consistent support at the rigid and VVE-addressed locations, while Refresh covers neither source. GT-conditioned diagnostics isolate routing from detection, whereas Dyn., Nxt., Direct MAVE, and DSR retain end-to-end evaluation. The supplement details the target construction.

Visual analysis. Figure 6 compares zero-motion reading with VVE backtracing. Red-to-green changes show displaced targets recovering same-class history. Because queried targets are fixed, the comparison isolates VVE address quality from the subsequent routing decision. On identical GT dynamic support, Figure 7 shows Persist following aligned support, Transport following displacement, and Refresh rejecting history. Fixing support isolates source choice from detection. This view complements the address control in Figure 6.

Together, the two visualizations summarize RoadOcc’s evidence-routing principle. VVE resolves *where* each queried voxel should retrieve historical evidence, while P/T/R verifies its semantic validity, preserving supported history and refreshing unreliable evidence before VDSF fusion.

5 CONCLUSION

In this work, we present RoadOcc, a voxel-level evidence-source routing framework for temporal roadside occupancy. Rather than relying on implicit temporal aggregation, RoadOcc explicitly determines where supported history should be retrieved and whether it should be reused. DCA uses current image evidence to identify motion-sensitive voxels, VVE estimates their historical source addresses, and VDSF turns these addresses into Persist, Transport, or Refresh decisions. On the reported InfraOcc protocol, the unified formulation achieves state-of-the-art occupancy accuracy while improving dynamic recovery and velocity estimation. Future work will extend this principle to moving platforms, longer temporal horizons, and instance-level modeling.

Table 5: **Controlled address and source selection.** All variants use the same sparse-token budget and model capacity.

Variant	Design		Output			
	VVE	P/T/R	Dyn.↑	Nxt.↑	dM↓	DSR↑
Rigid read			30.10	29.44	2.146	57.84
VVE read	✓		30.97	30.78	1.690	59.35
Full	✓	✓	32.37	31.72	1.669	61.11

Table 6: **P/T/R routing controls.** Mean $\pm$ standard deviation over three independent seeds.

Variant	Dyn.↑	Nxt.↑	DSR↑	dM↓
None	30.97 $\pm$.27	30.78 $\pm$.34	59.35 $\pm$.39	1.690 $\pm$.019
Gate	31.72 $\pm$.11	31.08 $\pm$.79	59.88 $\pm$.82	1.681 $\pm$.023
State	32.10 $\pm$.34	31.43 $\pm$.32	60.62 $\pm$.35	1.674 $\pm$.017
Full	32.37$\pm$.11	31.72$\pm$.26	61.11$\pm$.38	1.669$\pm$.020

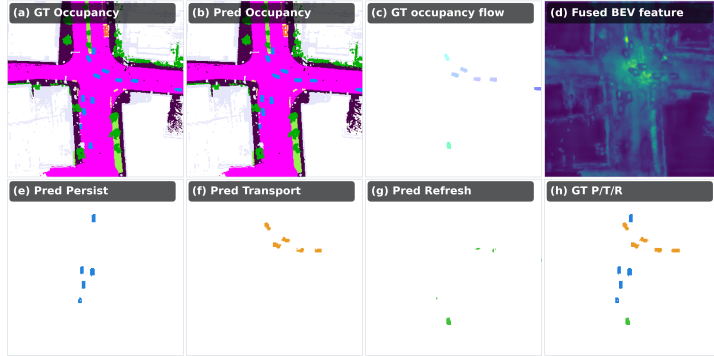


Figure 7: **Predicted P/T/R on GT dynamic support.**

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APPENDIX

This supplement documents the implementation and controlled evidence underlying RoadOcc. (1) Sections A–A.2 define the fixed-roadside data construction and velocity-target contract that enable VVE to estimate a class-consistent historical address. (2) Sections C–C.1 specify how P/T/R decides whether that addressed source is retained, transported, or refreshed before temporal fusion. (3) Sections B–F connect these two decisions to occupancy, motion, coverage, routing, and capacity diagnostics. (4) All controlled results retain the training configuration of the complete model in the main paper.

A ROADSIDE SENSOR CONFIGURATION

InfraOcc reorganizes synchronized V2X-Real streams from object-level cooperative perception into dense roadside occupancy supervision (Yang et al., 2026). Two fixed infrastructure units each carry one LiDAR and two high-resolution cameras under GPS time synchronization. Calibration maps the four camera streams, two point clouds, and occupancy annotations into a common roadside coordinate system. The four cameras are the only exteroceptive model inputs. LiDAR is used solely for benchmark construction and ground-truth generation. Thus the reported setting remains camera-only despite the multimodal data-collection platform.

Roadside benchmarks such as Rope3D and DAIR-V2X also study fixed or cooperative perception (Ye et al., 2022; Yu et al., 2022), but primarily evaluate object-level detection. InfraOcc instead retains a fixed coordinate system for dense occupancy, exposing voxel-level persistence and displacement. This property is essential for testing whether a current voxel retrieves its historical feature from a source-consistent location.

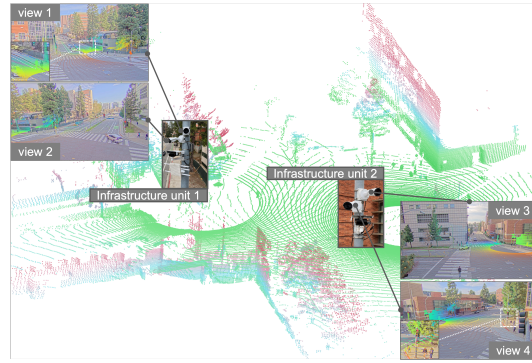


Figure A1: Infrastructure-side sensor configuration. Two fixed units provide four synchronized cameras and two LiDARs calibrated to a common roadside frame.

A.1 BENCHMARK SCOPE AND CONSTRUCTION CONTRACT

InfraOcc is a concurrent benchmark derived from V2X-Real. We are unaware of a second real-world benchmark that simultaneously provides fixed roadside sensing, temporally continuous dense semantic voxel labels, and a compatible camera-only evaluation protocol. Ego-centric occupancy benchmarks assume a moving sensor frame, while existing roadside and cooperative datasets predominantly expose object-level detection or tracking labels. Converting either setting into a second benchmark would require non-standard pseudo-labels and would not provide a controlled cross-dataset comparison. We therefore evaluate RoadOcc under one InfraOcc protocol and test architectural generality by transferring P/T/R to two temporal occupancy backbones under matched conditions.

The benchmark contains 290 temporally continuous roadside sequences, with 215 used for training and 75 for evaluation. Each sequence spans approximately 10–24 seconds. Raw streams are captured at 10 Hz and dense occupancy is labeled at 2 Hz, giving the 0.5 s interval used by RoadOcc. Each keyframe contains four calibrated camera views and a semantic occupancy volume over $[-64, 64] \times [-64, 64] \times [-4.8, 1.6]$ m at 0.4 m resolution, or $320 \times 320 \times 16$ voxels. Labels distinguish occupied, free, and unobserved space; unobserved voxels are excluded from supervision and evaluation.

Occupancy construction separates persistent structure from transient objects. A semantic static scaffold is accumulated from registered vehicle-side LiDAR after dynamic-point removal. Dynamic geometry is reconstructed from infrastructure-side LiDAR using annotated object tracklets, box-guided alignment, and local registration. The two sources are recomposed at each keyframe in the

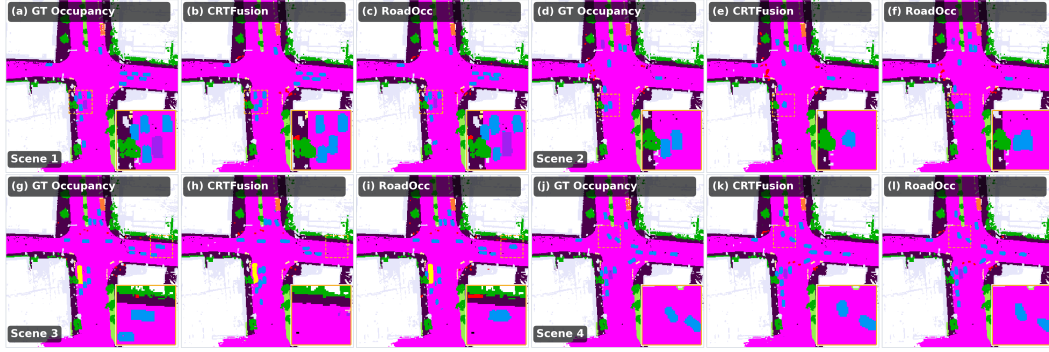


Figure A2: Matched-token occupancy comparison across four scenes. Each triplet shows GT, CRT-Fusion, and RoadOcc in a common BEV frame and semantic palette. The lower-right insets enlarge a representative region that RoadOcc recovers but CRT-Fusion mispredicts.

fixed roadside frame and voxelized. Ray traversal labels free space before the first observed surface, while voxels behind surfaces or outside valid rays remain unobserved. These LiDAR sources construct supervision only; RoadOcc receives no LiDAR feature as model input.

Qualitative occupancy recovery. Figure A2 compares both methods on identical tokens and coordinates. In the highlighted regions, CRT-Fusion either omits a participant or assigns an incorrect footprint, whereas RoadOcc more closely follows the GT extent and class. The surrounding road structure remains visually unchanged. The gain is therefore concentrated on transient participants rather than obtained by altering the dominant static background.

A.2 VELOCITY-TARGET CONSTRUCTION

InfraOcc occupancy labels do not by themselves define the transport field used by RoadOcc. We therefore generate planar voxel-velocity targets from annotated dynamic boxes and their temporal tracklets. Raw box velocity is transformed to the fixed roadside frame and may be replaced by the adjacent-frame tracklet displacement when the latter is valid and consistent: replacement is allowed when the raw estimate is invalid, the speed difference is at most 10 m/s, or the speed ratio is at most two. Speeds below 0.8 m/s are treated as stationary, and abnormal magnitudes are clipped to 60 m/s.

For track j , let $\mathbf{R}_{\text{rs} \leftarrow \text{raw}}$ transform planar vectors from the annotation frame into the fixed roadside frame, and let $\mathbf{c}_{j,t}^{\text{rs}}$ be its roadside-frame center. The two candidate velocities are

$$\tilde{\mathbf{v}}_{j,t} = \mathbf{R}_{\text{rs} \leftarrow \text{raw}} \mathbf{v}_{j,t}^{\text{raw}}, \quad \mathbf{d}_{j,t} = \frac{\mathbf{c}_{j,t+\Delta t}^{\text{rs}} - \mathbf{c}_{j,t}^{\text{rs}}}{\Delta t}, \quad \Delta t = 0.5 \text{ s}. \quad (11)$$

We denote the selected candidate by $\bar{\mathbf{v}}_{j,t}$. A valid tracklet displacement replaces the raw estimate when the raw estimate is invalid, when their speeds differ by at most 10 m/s, or when their speed ratio is at most two. The corrected velocity is rasterized over dynamic occupancy inside each box, whose vertical extent is expanded by a factor of 1.5 to absorb sparse boundary misalignment. Let $\tilde{B}_{j,t}$ denote this expanded box. The native target assigned to a dynamic voxel is

$$\mathbf{v}_t^{\text{gt}}(\mathbf{x}) = \begin{cases} \mathbf{0}, & \|\bar{\mathbf{v}}_{j,t}\|_2 < 0.8 \text{ m/s}, \\ \text{clip}_{60}(\bar{\mathbf{v}}_{j,t}), & \mathbf{x} \in \tilde{B}_{j,t} \cap \mathcal{O}_t^{\text{dyn}}, \end{cases} \quad (12)$$

where clip_{60} limits the magnitude to 60 m/s. For candidates no faster than 2 m/s, a one-step (0.5 s) future-occupancy check compares zero-motion and transported same-class support. The check is applied only with at least eight support voxels; zero motion is selected when its score exceeds the transported score by at least 0.05. Finally, face-connected component repair propagates valid motion within the corresponding raw dynamic occupancy component, and targets are generated at native, 1/2, 1/4, and 1/8 scales.

Future occupancy is used only to construct offline supervision and evaluation targets. Neither future images nor future features are provided to RoadOcc at training or inference time. The P/T/R threshold $\epsilon = 10^{-3}$ m/s in the main paper is a numerical zero test applied after this construction;

the physical stationary decision in target generation uses the 0.8 m/s threshold above. Stating both thresholds separates label denoising from the executed P/T/R routing rule.

B EVALUATION CONTRACT

Aggregate occupancy is dominated by persistent structure and can therefore conceal missed or stale traffic participants. Standard protocols emphasize semantic and geometric IoU (Wang et al., 2023; Tian et al., 2023), whereas dynamic occupancy and flow require motion-aware diagnostics (Ma et al., 2024; Liu et al., 2024). Table A1 formalizes our evaluation contract on InfraOcc (Xiang et al., 2024; Yang et al., 2026). Dynamic and next-frame IoU measure semantic recovery at the current and flow-advected next-frame locations. Direct MAVE and DSR jointly measure motion accuracy and coverage. Static, geometric, and ray metrics serve as guardrails against degrading the fixed scene. This coupled evaluation prevents a model from appearing favorable merely by predicting fewer dynamic voxels or retaining stale evidence.

Table A1: Metric and diagnostic scope.

Metric	Support	Failure	Role
Dyn. mIoU $\uparrow$	Dynamic GT voxels	Omission / class error	Dynamic semantic recovery
Static / gIoU $\uparrow$	Static / occupied support	Static-structure damage	Static-scene guardrail
Next Dyn. $\uparrow$	Advected dynamic GT	Displacement error	Next-frame consistency
Direct MAVE $\downarrow$	All dynamic GT voxels	Miss-induced motion bias	Unconditional flow error
TP-MAVE $\downarrow$ / DSR $\uparrow$	Matched semantics / dynamic GT	Conditional-match bias	Matched accuracy / coverage
RayIoU $\uparrow$	Visible evaluation rays	Surface-depth mismatch	Geometric fidelity

Direct MAVE evaluates every valid GT dynamic voxel, whereas TP-MAVE is conditional on a correct semantic match and must therefore be interpreted with DSR. A valid routing improvement should increase dynamic IoU and DSR, decrease Direct MAVE, and preserve the static and geometric guardrails. Reporting these quantities together distinguishes improved semantic correspondence from a lower conditional error caused by reduced semantic support. For completeness, let $\mathcal{D}$ be the set of evaluated dynamic classes, $V_c = \{i : y_i = c\}$ the valid GT voxels of class c , and $V_c^{\text{TP}} = \{i \in V_c : \hat{y}_i = c\}$. Classes without valid support are omitted from the corresponding class macro-average; $\mathcal{D}'$ and $\mathcal{D}''$ denote the supported classes for Direct and TP-MAVE, respectively. The two velocity metrics and semantic recall are

$$\text{Direct MAVE} = \frac{1}{|\mathcal{D}'|} \sum_{c \in \mathcal{D}'} \frac{1}{|V_c|} \sum_{i \in V_c} \|\hat{\mathbf{v}}_i - \mathbf{v}_i\|_2, \quad (13)$$

$$\text{TP-MAVE} = \frac{1}{|\mathcal{D}''|} \sum_{c \in \mathcal{D}''} \frac{1}{|V_c^{\text{TP}}|} \sum_{i \in V_c^{\text{TP}}} \|\hat{\mathbf{v}}_i - \mathbf{v}_i\|_2. \quad (14)$$

Thus MAVE is class-macro while DSR is a micro recall over GT dynamic voxels. For next-frame dynamic IoU, centers of predicted current dynamic voxels are advanced by the predicted planar velocity for 0.5 s, transformed to the next roadside frame, and voxelized. Collisions use deterministic per-class majority assignment before class-macro IoU is computed against next-frame GT. This flow-advected diagnostic is distinct from the same-class radius-one hit score used only in the backwarp visualization.

$$\text{DSR} = \frac{\sum_{c \in \mathcal{D}} |V_c^{\text{TP}}|}{\sum_{c \in \mathcal{D}} |V_c|}. \quad (15)$$

C TRAINING OBJECTIVES

RoadOcc groups its objectives by function so that each loss supervises one decision in the DCA-VVE-VDSF chain. Camera-depth supervision uses weight 0.5. Semantic occupancy is supervised from full resolution to 1/8 scale with weights [1.0, 0.5, 0.25, 0.125]. The static, dynamic, and DCA-candidate auxiliaries each use weight 0.5. VVE receives class- and speed-balanced robust flow supervision at all four scales with weight 0.1. Its local-connectivity term also uses weight 0.1 and is enabled after epoch 3, which prevents noisy early correspondences from dominating optimization.

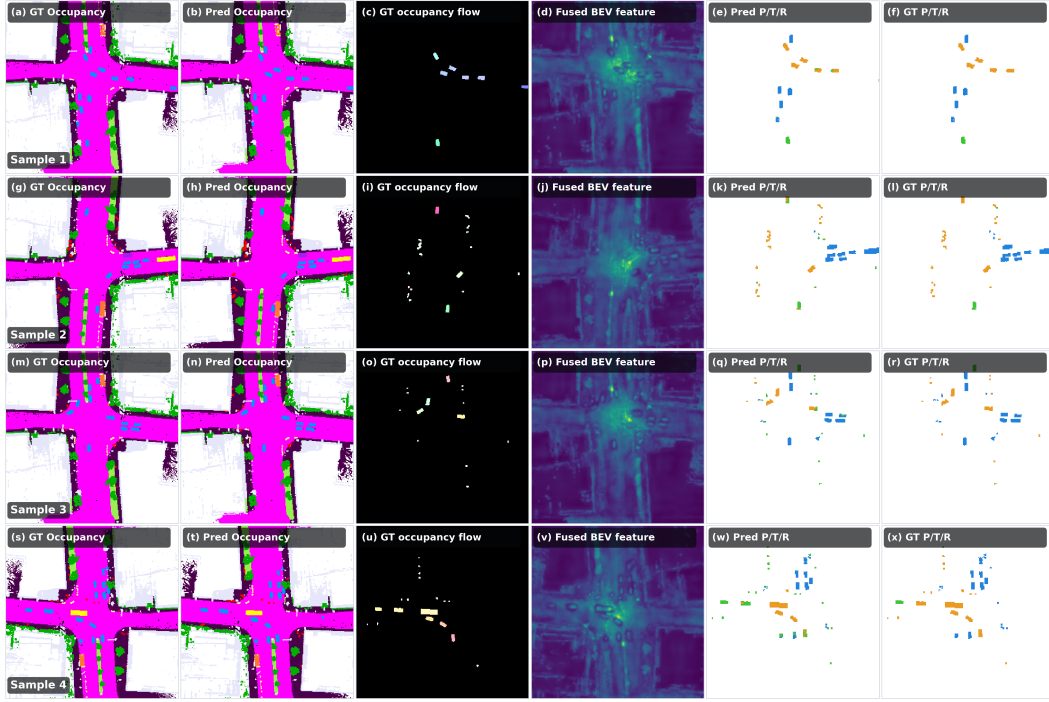


Figure A3: Native-scale P/T/R routing across four samples. Predicted and GT routes share GT dynamic support. Blue, orange, and green denote Persist, Transport, and Refresh, respectively.

P/T/R is supervised at native, $1/2$, $1/4$, and $1/8$ scale with relative weights $[1, 0.5, 0.25, 0.125]$, normalized across scales, and total weight $\lambda_{\text{route}} = 0.5$. Each scale uses its physical cell size to construct the same semantic-support target. Cross-entropy uses equal P/T/R class weights; from epoch 2, a soft-overlap term of weight 0.5 uses state weights $[1, 1, 0.25]$. This coarse-to-fine supervision aligns every executed routing field with the resolution at which it controls temporal fusion.

C.1 CANDIDATE, MOTION, AND ROUTING CONTRACT

DCA score. Let π_{cur} and π_{sta} denote current and static-reference class probabilities, and let $\mathcal{C}_{\text{sta}}^+$ contain the static classes together with the empty class. The executed change and dynamic scores are

$$\delta(\mathbf{x}) = \frac{1}{|\mathcal{C}_{\text{sta}}^+|} \sum_{c \in \mathcal{C}_{\text{sta}}^+} |\pi_{\text{cur},c}(\mathbf{x}) - \pi_{\text{sta},c}(\mathbf{x})|, \quad (16)$$

$$p_{\text{dyn}}(\mathbf{x}) = \sum_{c \in \mathcal{C}_{\text{dyn}}} \pi_{\text{cur},c}(\mathbf{x}), \quad (17)$$

$$c(\mathbf{x}) = \frac{\max\{\delta(\mathbf{x}), p_{\text{dyn}}(\mathbf{x})\}}{\max_{\mathbf{y}} \max\{\delta(\mathbf{y}), p_{\text{dyn}}(\mathbf{y})\} + 10^{-6}}. \quad (18)$$

DCA ranks this normalized score. VDSF combines it with nonempty support before the fixed top- K selection. For an anchor $\mathbf{r} = (\mathbf{x}, z_r)$, $c(\mathbf{r})$ is the score of the corresponding voxel-height bin and is shared across cameras before visibility masking. Let $d_{n,r}$ be its projected depth, $\hat{d}_n$ the winning predicted depth bin, and $\Delta_d = 1$ m the bin width. The executed depth-consistency factor is

$$\beta_{n,r} = \exp \left[-\frac{\sigma^2}{2} \min_{\xi \in \{-\Delta_d, +\Delta_d\}} |d_{n,r} - (\hat{d}_n + \xi)|^2 \right], \quad \sigma = 2. \quad (19)$$

Deformable-attention weights are first softmax-normalized over sampled points, then multiplied by β . They are not renormalized afterward; camera outputs are averaged over cameras retaining at least one visible anchor.

VVE correspondence range. At scale s , the temporal branch forms a 5×5 correlation volume with radius $r = 2$. With a native 0.4 m grid and a 0.5 s interval, the explicit correlation evidence spans the following physical ranges.

Each correlation operand is normalized to unit ℓ_2 norm with $\epsilon = 10^{-6}$. The resulting volume has 25 channels and is concatenated with current and historical BEV embeddings, processed by two 3×3 convolution blocks, and decoded as a two-dimensional velocity residual in m/s. The window bounds only the explicit matching evidence. The coarse velocity is added to each finer prediction and the final continuous residual is not clipped, so it does not impose a hard bound on predicted speed. The GT-flow study in the main paper isolates the value of transport addresses across speed ranges. In the temporal blend, the history branch receives weight $0.5 \mathbf{1}_{\text{hist}} \max_z p_{\text{dyn}}(\mathbf{x}, z)$; this is current dynamic support gated by history validity, rather than a calibrated measure of historical correspondence.

Native P/T/R target. For a current dynamic voxel, the Transport source is

$$\mathbf{x}_{\text{src}} = \text{round}\left(\mathbf{x} - \frac{\Delta t \mathbf{v}_t(\mathbf{x})}{0.4 \text{ m}}\right), \quad \Delta t = 0.5 \text{ s.} \quad (20)$$

Persist tests the rigid address and Transport tests $\mathbf{x}_{\text{src}}$. Support means that the nearest previous GT contains the same semantic class at the same height within a 3×3 XY neighborhood (L_∞ radius one). Support is Boolean: one or more matching voxels suffice, with no instance-level tie-breaking. Out-of-grid or unsupported candidates are assigned Refresh rather than removed. The target follows InfraOcc’s voxel-level semantic supervision and therefore tests class-consistent support without requiring instance identities.

Coarse-to-fine FIFO routing. At each decoder scale, the shared controller independently projects the current feature and nearest rigidly aligned history to eight channels. It adds eight-channel projections of the detached VVE flow (normalized by 10 m/s) and the detached, upsampled coarser P/T/R distribution, then applies a $3 \times 3 \times 3$ refinement and a three-logit classifier. A lightweight native head upsamples the 1/2 context and combines it with native flow and the finest P/T/R distribution to produce the full-resolution endpoint. Each execution-scale distribution is shared across valid FIFO slots; slot age and cached VVE fields define source-consistent inverse-trajectory Transport addresses. Routed slots are concatenated before the fusion MLP. Invalid slots are masked, and every scene boundary clears all stage-local queues. Fusion uses the soft distribution; argmax is reserved for diagnostics.

D CAPACITY AND EFFICIENCY CONTEXT

Table A3 reports parameters, end-to-end latency, and occupancy accuracy for reproduced local configurations. The table provides a common camera-only efficiency context. The main paper evaluates the RoadOcc routing mechanism under its matched temporal protocol.

Class-wise improvement. Figure A4 resolves the official result in Table 2 into dynamic and static classes. Relative to STCOcc, RoadOcc improves five of the six dynamic classes, led by Bike at +10.47 and Bus at +6.68, while Pedestrian changes by -0.19 . Every static class improves. This paired behavior shows that source-consistent routing recovers transient participants without weakening persistent roadside structure. The category pattern follows the aggregate dynamic-static split. Gains on Bike, Bus, Car, Motorcycle, and Truck support the intended recovery of moving participants. The widespread static gains show that corrected temporal addresses and source selection preserve the persistent scene rather than trading it for dynamic accuracy.

Table A2: Physical support of the local VVE correlation window.

Scale	Cell	Disp.	Speed equiv.
1/8	3.2 m	± 6.4 m	± 12.8 m/s
1/4	1.6 m	± 3.2 m	± 6.4 m/s
1/2	0.8 m	± 1.6 m	± 3.2 m/s

Table A3: Accuracy–efficiency comparison.

Setting	Params.↓	Lat.↓	All↑	Dyn.↑
BEVDet ’21	100.66	41.92	45.14	21.19
BEVFormer ’22	69.21	180.59	47.76	10.73
BEVDepth ’23	100.66	57.43	46.81	19.64
TPVFormer ’23	80.76	134.63	43.09	4.36
SurroundOcc ’23	80.04	44.40	50.77	11.37
CONet ’23	100.74	53.23	49.25	18.93
SparseOcc ’24	105.08	64.30	45.42	9.41
GaussianFormer ’24	36.32	65.50	43.63	10.45
RoadOcc (ours)	78.69	TBD	65.29	32.37

E CONTROLLED DESIGN STUDIES

All studies hold the backbone, data protocol, optimizer, training schedule, scene-boundary queue reset, and VDSF configuration fixed. Unless varied explicitly, the decoder stages use $K = [2000, 512, 128]$ sparse tokens and retain $[8, 4, 2]$ history samples, respectively. The main paper reports the matched address and P/T/R controls. The supplementary studies below provide non-overlapping evidence on VVE training, route prediction, sparse allocation, and address construction.

VVE training design. The main-paper control establishes the value of reading from a corrected address. Figure A5 follows a progressive training sequence that introduces multi-scale flow supervision, candidate-map supervision, connectivity regularization, and flow-addressed history retrieval.

Progressive training behavior. Dyn. improves monotonically while Direct MAVE decreases as the VVE signals are introduced. Across the full progression, Dyn. rises by +3.63 and Direct MAVE falls by 0.271. The final flow-addressed read contributes another +1.65 Dyn. while Direct MAVE decreases by only 0.021, showing that its main benefit comes from retrieving useful history.

Route-wise diagnostics. The main paper establishes the causal value of P/T/R. Table A4 audits its source decisions on valid GT dynamic voxels, deliberately separating routing from occupancy detection. Persist and Transport account for 96.82% of the target support and retain high precision. Transport is more selective because it additionally requires displaced correspondence. Refresh is rare and therefore remains the most difficult state. Its higher predicted share and lower precision indicate conservative rejection of uncertain history, while its recall preserves unsupported cases. Because VDSF consumes soft route distributions, these argmax statistics diagnose route preference rather than replace the end-to-end Dyn., Nxt., Direct MAVE, and DSR evaluation.

Qualitative routing behavior. Figure A3 relates each route to observable scene evidence. Persist coincides with aligned objects, Transport follows objects with nonzero displacement, and Refresh appears where no supported historical source exists. Because predicted and GT routes are rendered on identical dynamic support, their differences reflect routing decisions rather than occupancy misses. The maps recover the dominant Persist and Transport structure, while the remaining discrepancies concentrate on the less frequent Refresh state. This visualization should therefore be read as a semantic source-choice audit, not as an instance-correspondence or detection metric.

Sparse-token budget. VDSF processes only the top- K selected voxels. The token study tests whether the reported behavior depends on a larger compute budget or on identifying the right locations. All other model and routing settings remain fixed while K is scaled at every decoder stage. Increasing the small allocation to the default improves Dyn. by 0.63 and DSR by 0.83. Doubling the default allocation changes Dyn. by only 0.02, dMAVE by 0.004, and DSR by 0.07, while next-frame

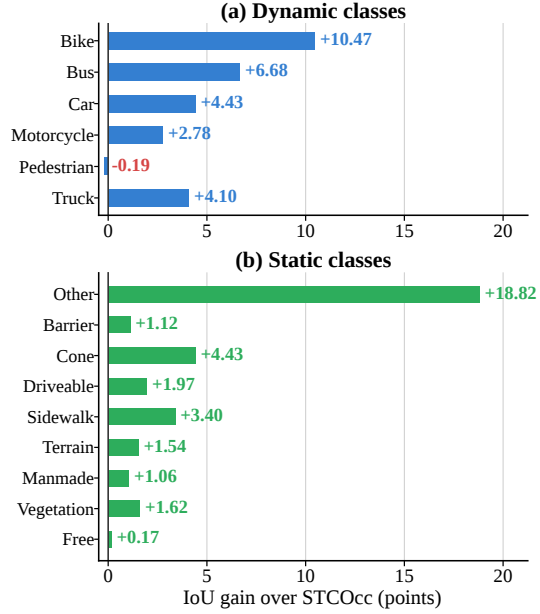


Figure A4: Class-wise IoU gains over STCOcc.

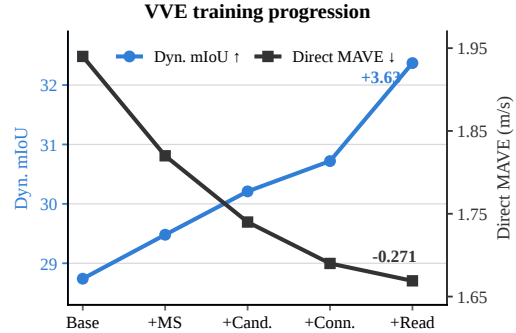


Figure A5: Dynamic occupancy and motion-error progression under the VVE training signals.

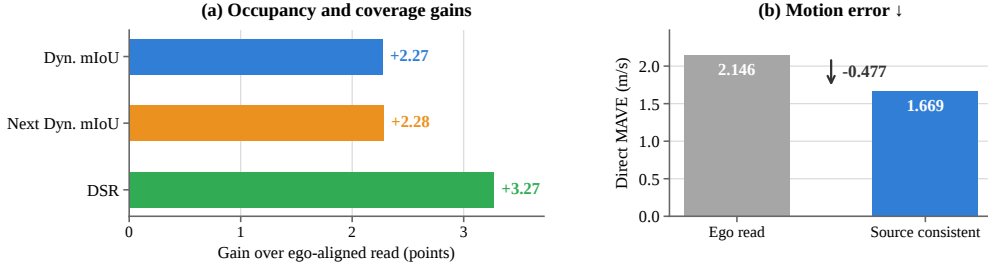


Figure A6: Measured advantage of source-consistent address construction over ego-aligned reading. Values are derived from the matched controlled settings in Table A6.

IoU fluctuates by -0.03 . This saturation shows that the reported result does not rely on continually expanding the sparse-token budget.

Velocity-address construction. Velocity is useful for fusion only when it points to evidence that explains the current target. Table A6 therefore compares a rigid ego-aligned read, a target-vector backtrace, and the complete source-consistent rule under an otherwise fixed P/T/R router. This isolates address construction from subsequent source acceptance and complements flow-based warping in prior occupancy models (Liu et al., 2024; Liao et al., 2025).

Target-vector backtracing improves Dyn. by 1.31 and reduces dMAVE by 0.442 over the ego-aligned read. Enforcing source consistency contributes a further 0.96 Dyn., 1.06 Nxt., and 1.29 DSR while reducing dMAVE by 0.035. The paired occupancy and coverage gains show that the refined address is useful for retrieval rather than only for velocity regression.

Stepwise address benefit. Figure A6 makes the controlled progression in Table A6 explicit. Target-vector tracing corrects motion-aware retrieval, while source consistency adds a further gain by ensuring that the transported source remains semantically admissible for the current voxel. Across the full progression, Dyn., Next Dyn., and DSR rise by 2.27, 2.28, and 3.27 points, respectively, while Direct MAVE falls by 0.477.

Table A4: Native-scale P/T/R routing statistics on valid GT dynamic voxels. Argmax diagnostics are percentages.

State	Target	Pred.	Precision	Recall	IoU
Persist	49.42	48.86	94.35	93.75	88.83
Transport	47.40	38.88	93.30	76.61	72.75
Refresh	3.19	12.26	18.65	71.41	17.29

Table A5: Sensitivity to sparse-token allocation.

Budget K	Dyn. $\uparrow$	Nxt. $\uparrow$	dM $\downarrow$	DSR $\uparrow$
[1000, 256, 64]	31.74	30.96	1.716	60.28
[2000, 512, 128]	32.37	31.72	1.669	61.11
[4000, 1024, 256]	32.39	31.69	1.665	61.18

Table A6: VVE address control.

Address	Dyn. $\uparrow$	Nxt. $\uparrow$	dM $\downarrow$	DSR $\uparrow$
Ego-aligned read	30.10	29.44	2.146	57.84
Target-vector trace	31.41	30.66	1.704	59.82
Source-consistent	32.37	31.72	1.669	61.11

F IMPLEMENTATION AND SCOPE

The production configuration uses four cameras, a ResNet-50 backbone, three decoder stages, a 0.4 m voxel grid, and a 0.5 s sampling interval. Models are optimized with AdamW for 24 epochs. Controlled variants share the same initialization policy, optimizer, schedule, scene-boundary queue reset, and InfraOcc evaluator. Only the designated factor is changed.

P/T/R operates on semantic voxel support, matching the InfraOcc label space. Nearby same-class objects can share support within the local radius; when history has no supported source, Refresh uses the current-frame feature. The route target therefore supervises semantic evidence admissibility rather than instance identity. We report dynamic IoU, Direct MAVE, and DSR jointly so that semantic recovery, motion error, and support coverage remain visible beyond the GT-conditioned route diagnostic.